Predictive Modeling with Classical Machine Learning

Minor Project Report - Artificial Intelligence Domain

Organization: InternVision Technology Private Limited

Abstract:

This report details the end-to-end development, implementation, and evaluation of a classical Machine Learning classification pipeline. The project serves as a foundational exercise in applying supervised learning algorithms to structured datasets. Utilizing the widely recognized Iris Classification Dataset, the project demonstrates proficiency in essential data science workflows, including environment configuration, data preprocessing, feature separation, model training, and rigorous performance evaluation. A Logistic Regression model was successfully deployed, achieving perfect classification metrics on unseen testing data.

1. Project Overview

In the field of Artificial Intelligence, the ability to accurately classify data based on historical patterns is a cornerstone of predictive modeling. The primary objective of this Minor Project was to establish foundational competencies in this area by solving a multi-class classification problem.

The project focuses heavily on the Machine Learning lifecycle. Rather than relying on complex, resource-heavy Deep Learning models, this exercise emphasizes the power and efficiency of classical algorithms. By building a linear model from scratch, the project highlights the importance of understanding underlying data structures and the mechanics of standard evaluation metrics.

2. Technologies and Environment Setup

To adhere to software engineering best practices, the project was developed within an isolated virtual environment. This approach prevents dependency conflicts and ensures that the project can be seamlessly reproduced on different operating systems.

- **Programming Language:** Python 3.10
- **Development Environment:** Anaconda Distribution, executed via Jupyter Notebook for interactive cell-by-cell execution.
- **Core Dependencies:**
    - pandas (Data Manipulation): Utilized for transforming raw arrays into highly structured DataFrames, allowing for efficient querying and data inspection.
    - scikit-learn (Machine Learning): The primary engine for the project, providing the built-in dataset, the algorithm implementations, and the standardized metric calculation functions.
    - matplotlib (Visualization): Included for potential graphical representation of data distributions.
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Environment Configuration Commands:

```
conda create -n internvision_ml python=3.10
conda activate internvision_ml
conda install pandas scikit-learn matplotlib jupyter
```

3. Dataset Description

The project utilizes the **Iris Classification Dataset**, a staple in the machine learning community originally introduced by the British statistician and biologist Ronald Fisher.

- **Total Volume:** 150 instances of Iris flowers.

- **Dimensionality:** Four continuous, numeric features representing physical dimensions in centimeters:
    - Sepal length
    - Sepal width
    - Petal length
    - Petal width
- **Target Classes:** Three distinct species of Iris flowers (Setosa, Versicolor, and Virginica), mathematically encoded as 0, 1, and 2 respectively.
- **Data Integrity:** An initial diagnostic analysis utilizing Pandas confirmed that the dataset is completely pristine, containing zero null values, missing data points, or structural anomalies, eliminating the need for complex imputation strategies.

4. Implementation Methodology

4.1 Data Preprocessing

The raw data arrays provided by scikit-learn were mapped into a structured pandas DataFrame. To create a cohesive view of the data, the target variable array was appended as a distinct column alongside the feature columns.

4.2 Feature and Target Separation

A critical step in supervised learning is ensuring the model only learns from the inputs and does not have access to the answers during training. The dataset was bifurcated into:

- **Feature Matrix (X):** Containing the 4 dimensions, maintaining a shape of (150, 4).
- **Target Vector (y):** Containing the species classifications, maintaining a shape of (150,).

4.3 Train-Test Split

To ensure an unbiased, objective evaluation of the model's predictive power, the data was strictly partitioned to prevent data leakage.

- **Training Set:** 70% of the data (105 samples) was reserved exclusively for fitting the model.
- **Testing Set:** 30% of the data (45 samples) was hidden from the model during training to act as unseen, future data.
- A random_state of 42 was applied to the splitting algorithm to guarantee that the random distribution of the data remains consistent across multiple executions.

4.4 Model Selection and Training

A **Logistic Regression** algorithm was selected for this task. Despite its name, Logistic Regression is a robust linear model for classification. It uses a logistic function to model a binary dependent variable, which scikit-learn adapts for multi-class problems using a One-vs-Rest (OvR) or multinomial scheme.

The core mathematical intuition relies on the sigmoid function to output a probability between 0 and 1:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

The model was instantiated with the hyperparameter max_iter=200. This deliberate adjustment ensures that the algorithm's underlying solver has a sufficient number of iterations to converge on the optimal weights without throwing optimization warnings. The model was then successfully fitted using the X_train and y_train datasets.

5. Results and Evaluation Metrics

The trained model was tasked with predicting the species of the 45 unseen testing samples. Because the target contains three classes rather than a simple binary outcome, all evaluation functions were configured to use a weighted average, which calculates metrics for each label and finds their average weighted by support (the number of true instances for each label).

The model yielded the following exceptional metrics:

- **Accuracy (1.0000):** The model correctly predicted the class for exactly 100% of the test samples.
- **Precision (1.0000):** Out of all the instances the model predicted as a certain species, 100% of them were actually that species (Zero False Positives).
- **Recall (1.0000):** Out of all the actual instances of a certain species in the test set, the model correctly identified 100% of them (Zero False Negatives).
- **F1-Score (1.0000):** The harmonic mean of precision and recall sits at a perfect 1.0, indicating flawless balance between the two metrics.

6. Conclusion and Future Scope

The Logistic Regression model achieved a perfect score across all evaluated metrics. This indicates that the model flawlessly classified all unseen data into their correct categories.

This perfect performance is a direct result of the dataset's inherent structure. The physical dimensions of the petals and sepals provide exceptionally clear mathematical boundaries between the three classes. Because the data is highly linearly separable, a foundational linear classifier like Logistic Regression is capable of drawing decision boundaries without error.

Future Scope: While this project successfully demonstrates a full machine learning lifecycle, future iterations could explore more complex, noisy datasets (such as text or image data) where non-linear models like Random Forests or Neural Networks would be required to handle overlapping class boundaries.